

JOHN WICK UNIVERSE DATASET
EDA AND REGULARIZATION REPORT

Dataset Shape: (100000, 12)

Target Variable: loyalty

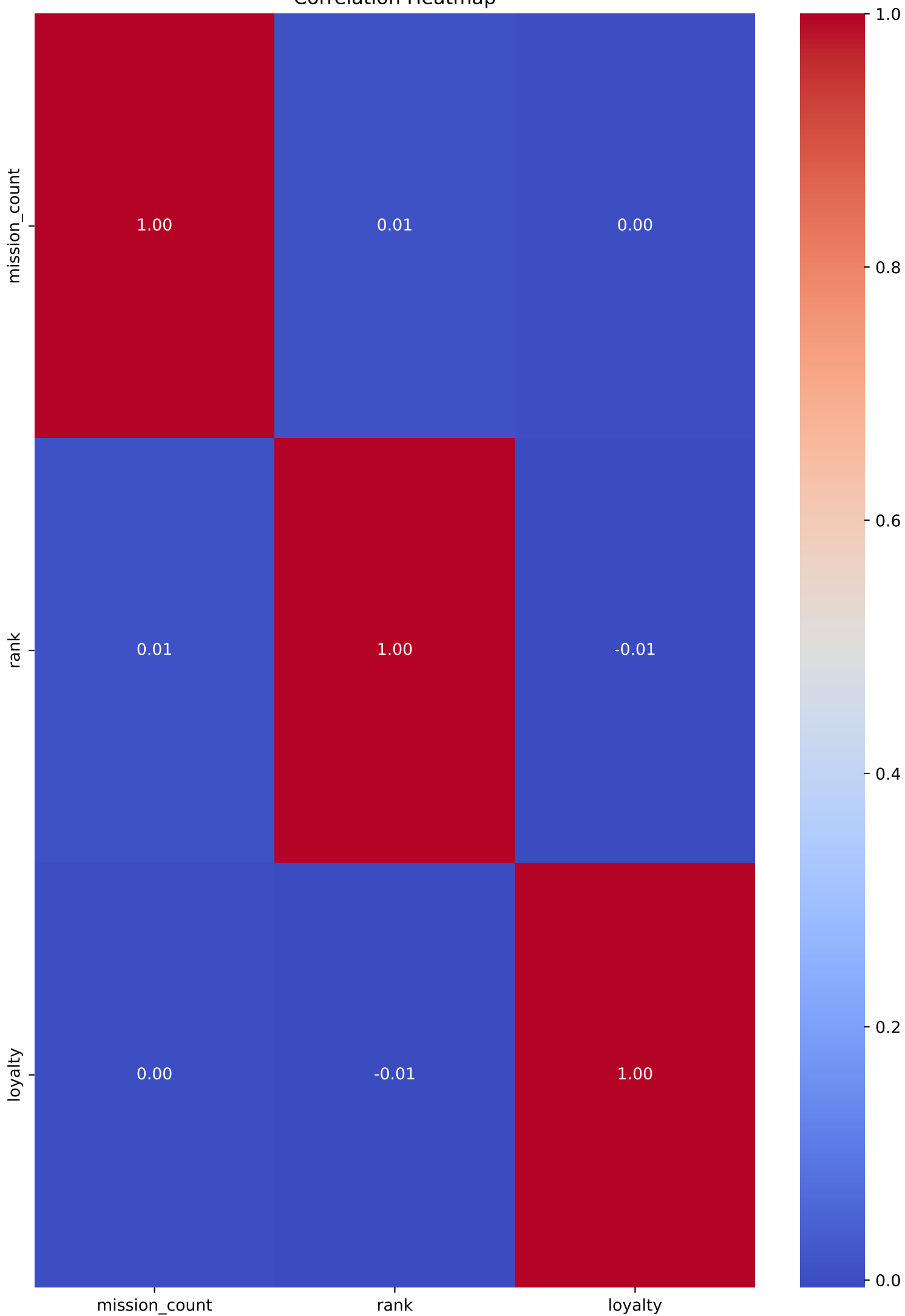
Numerical Features: 2
Categorical Features: 9

Missing Values: 0
Duplicate Rows: 4

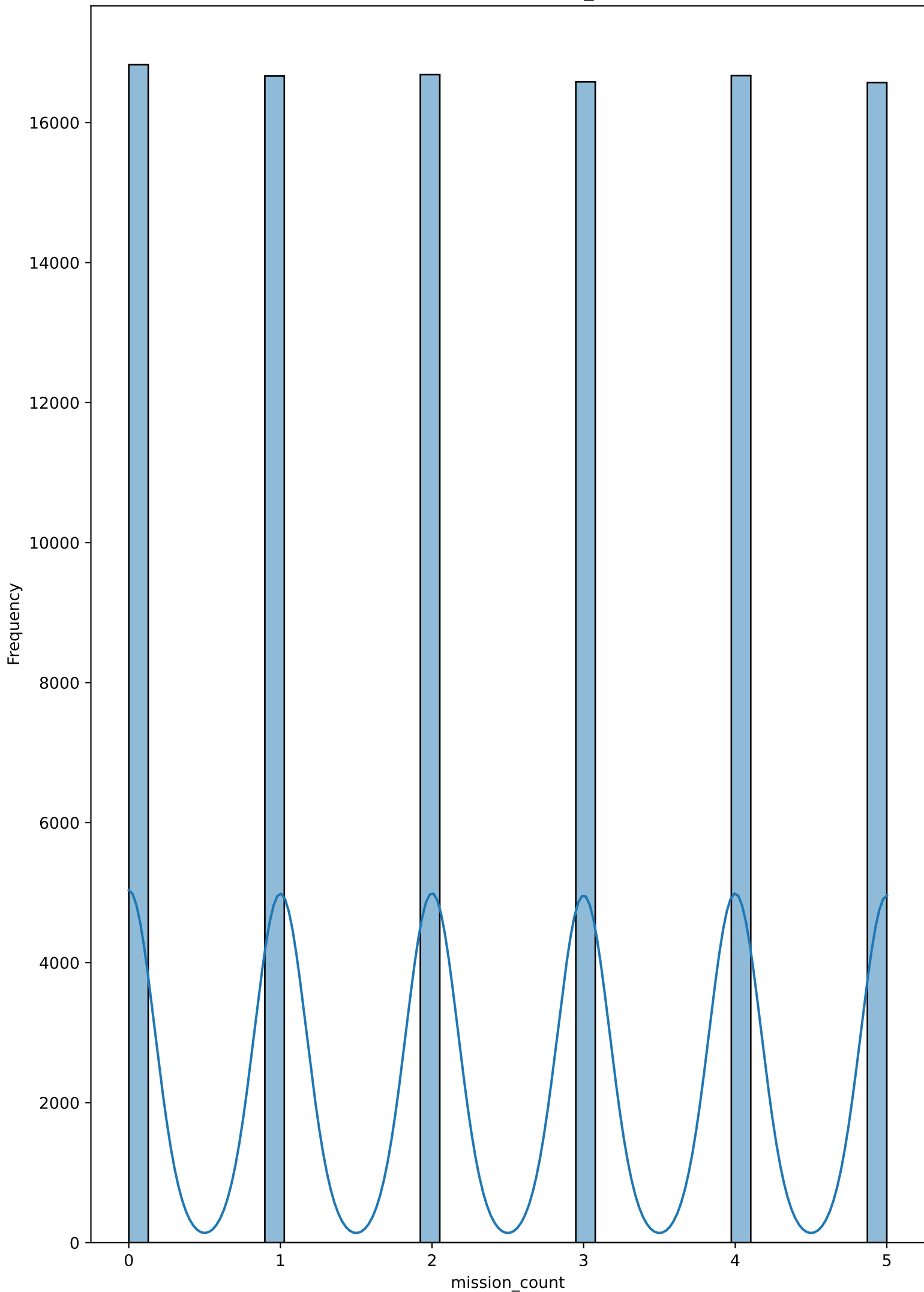
MODEL PERFORMANCE

	MAE	MSE	RMSE	R2 Score
Linear Regression	2.4974	8.2555	2.8732	-0.0018
Ridge Regression	2.4974	8.2555	2.8732	-0.0018
Lasso Regression	2.4977	8.2408	2.8707	-0.0000
Elastic Net	2.4977	8.2412	2.8708	-0.0001

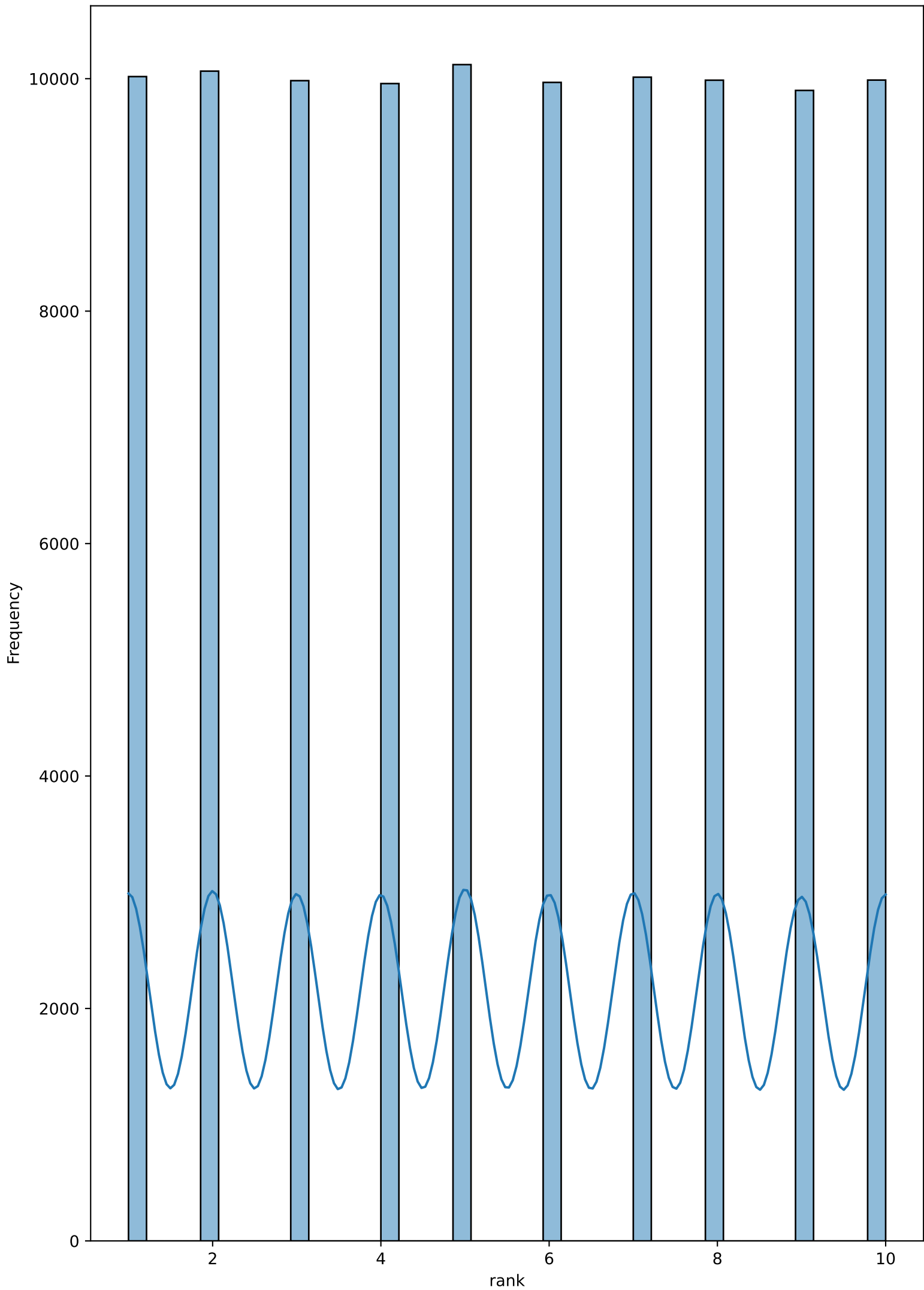
Correlation Heatmap



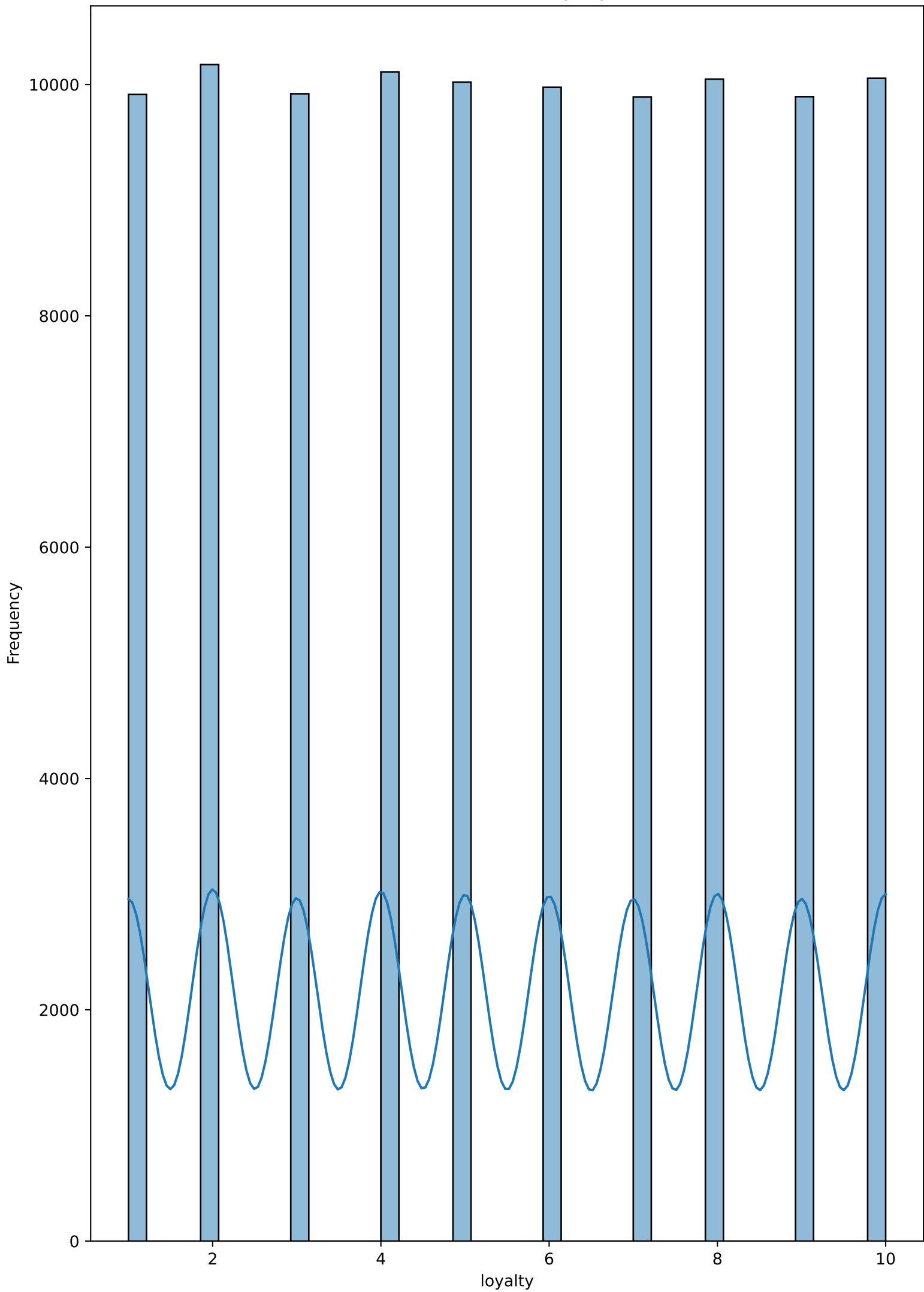
Distribution of mission_count



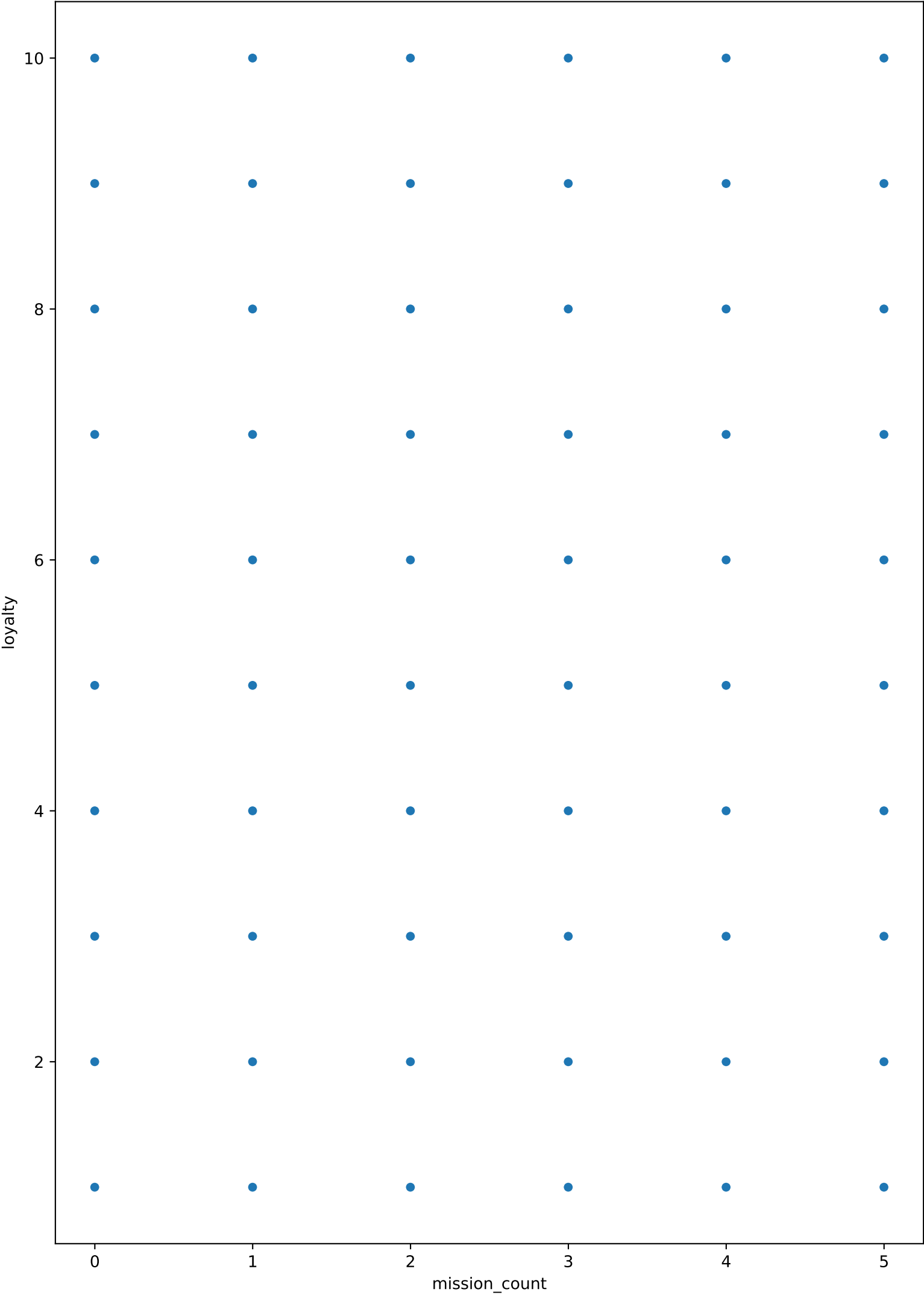
Distribution of rank



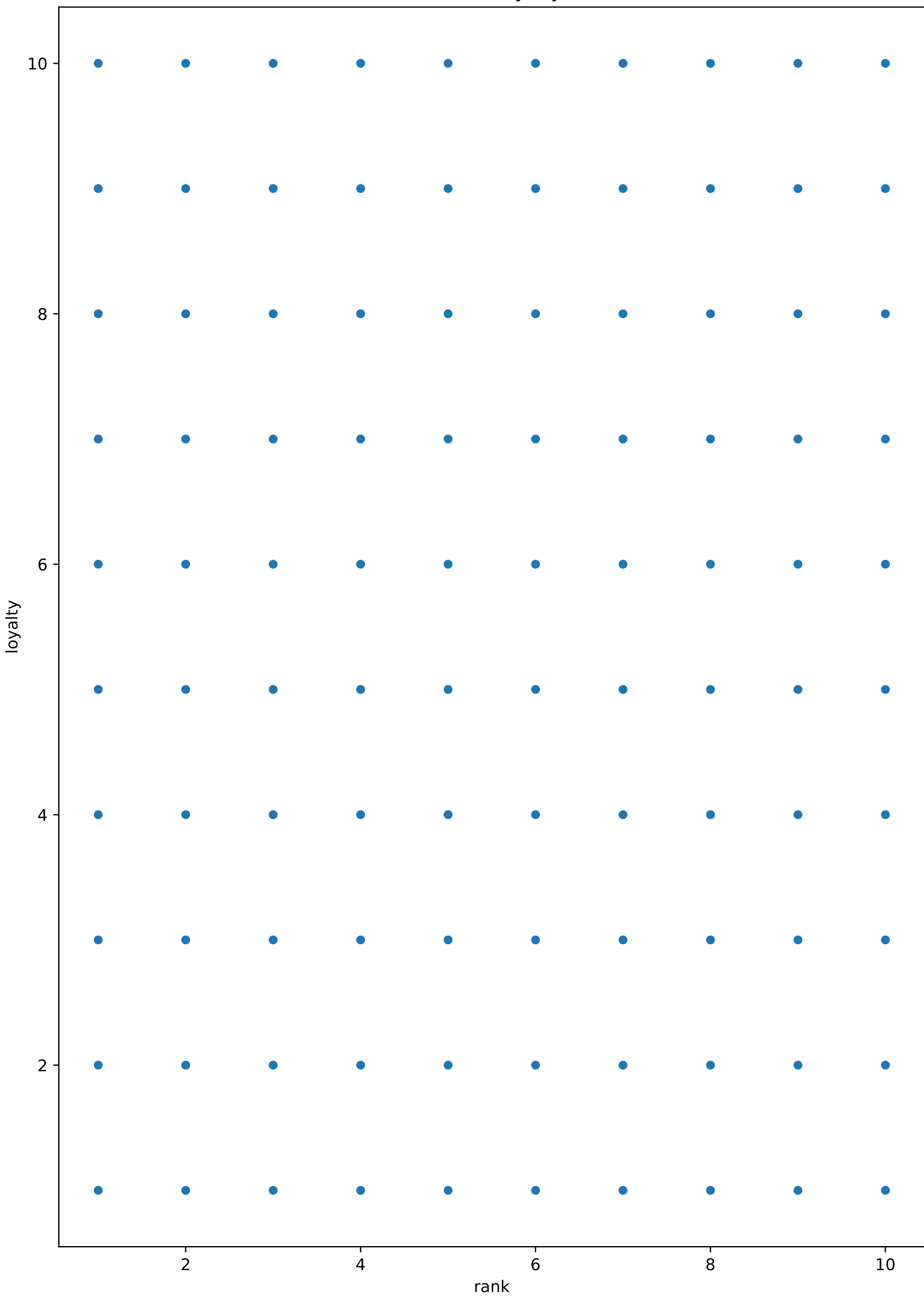
Distribution of loyalty

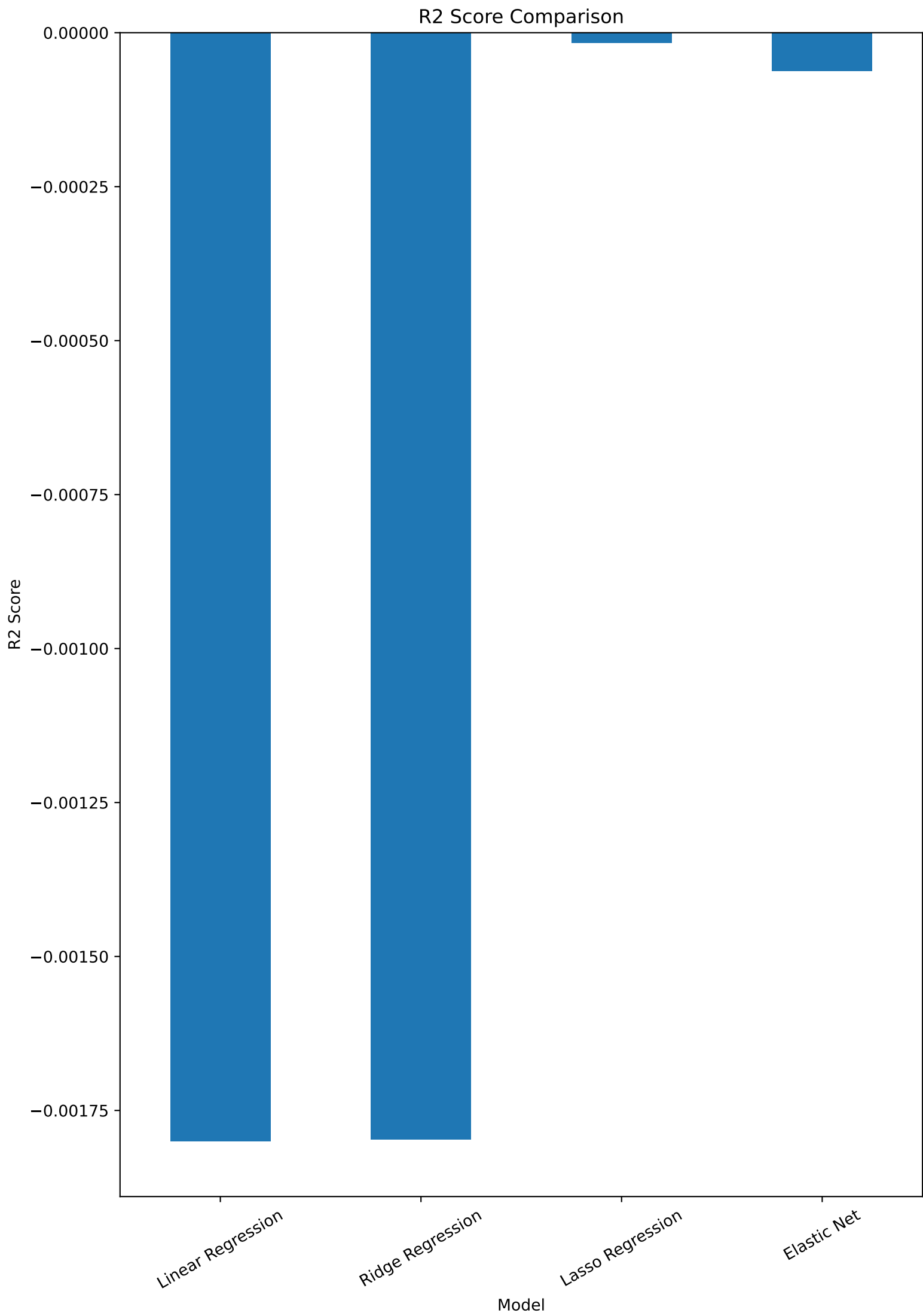


mission_count vs loyalty

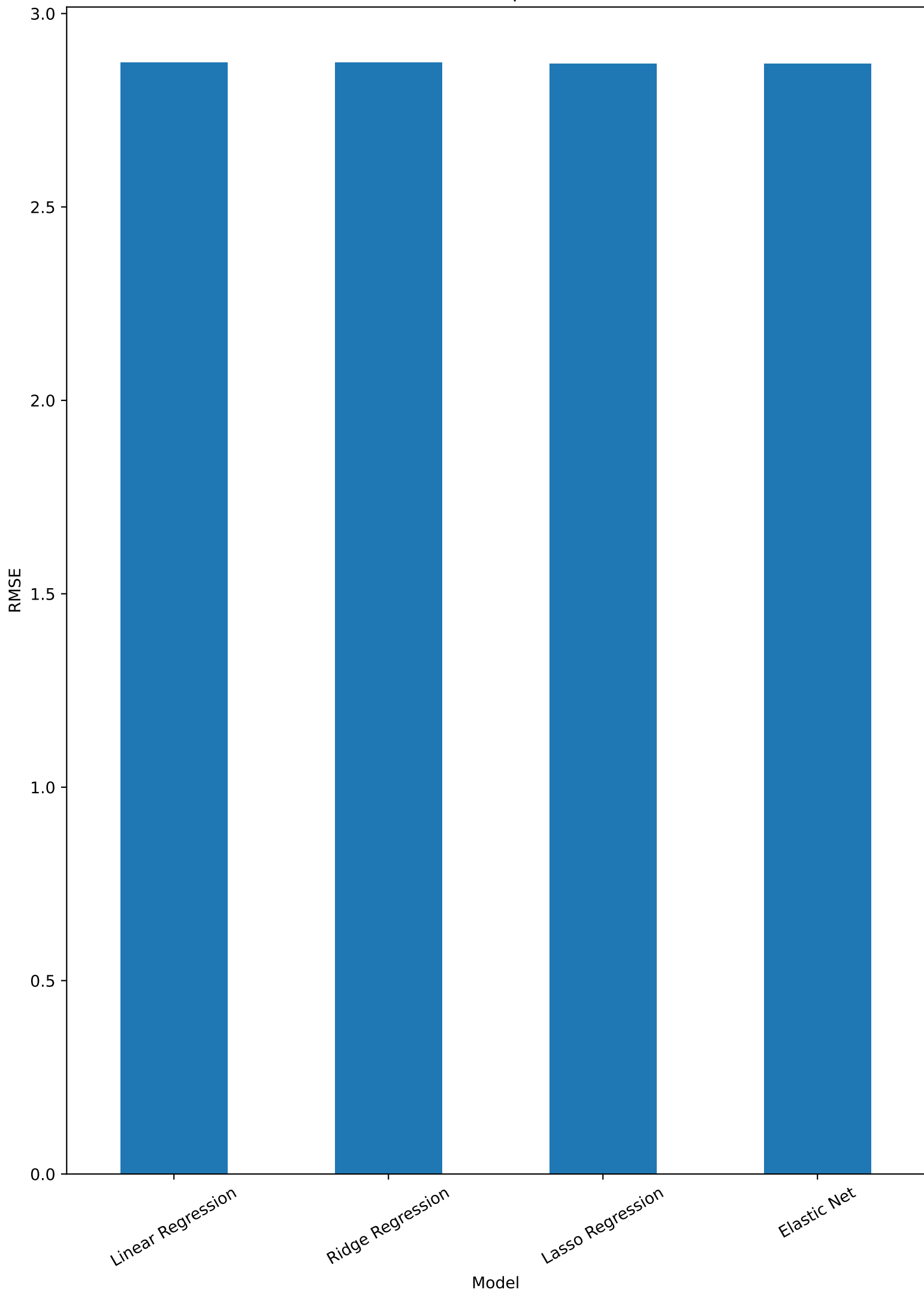


rank vs loyalty

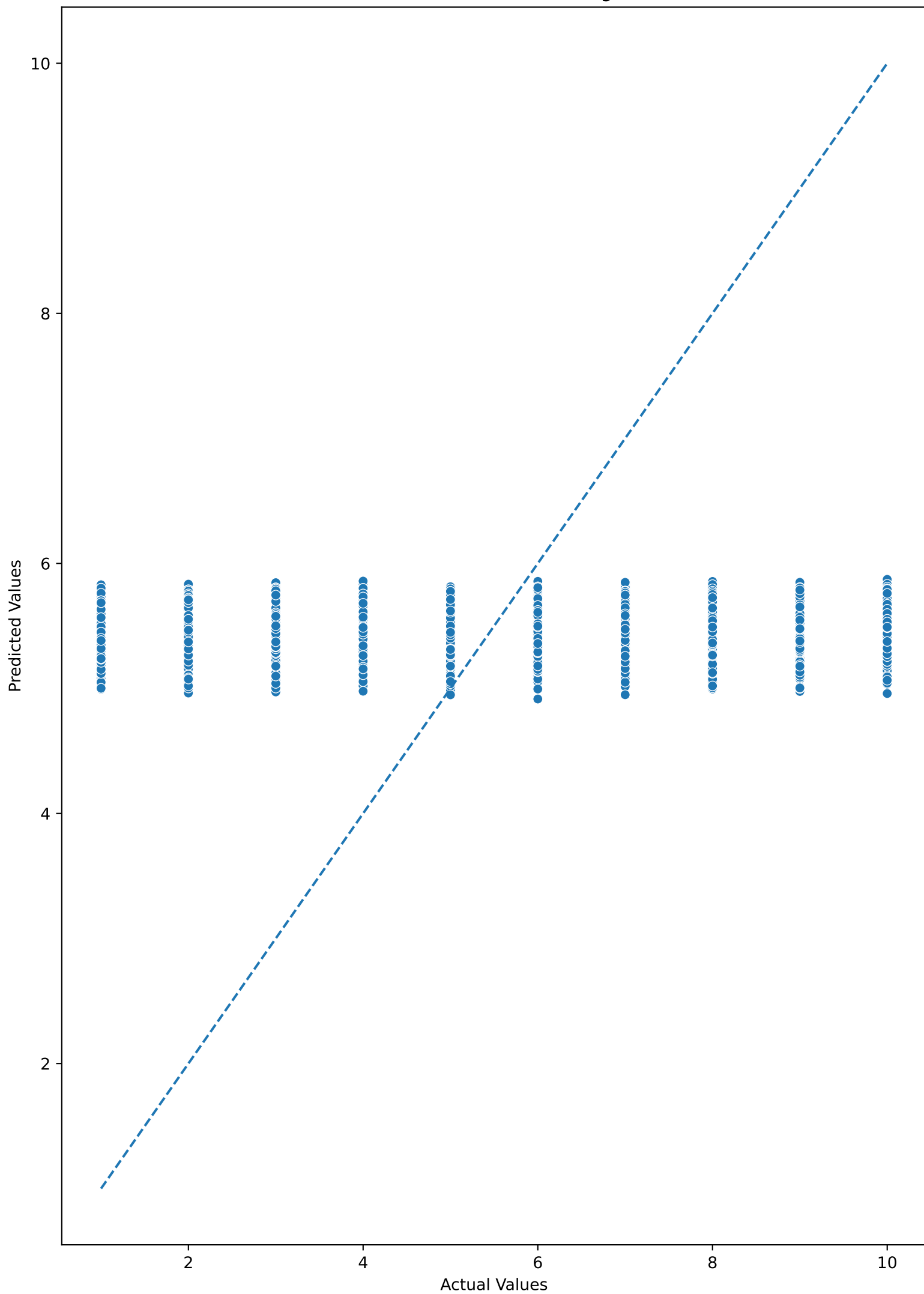




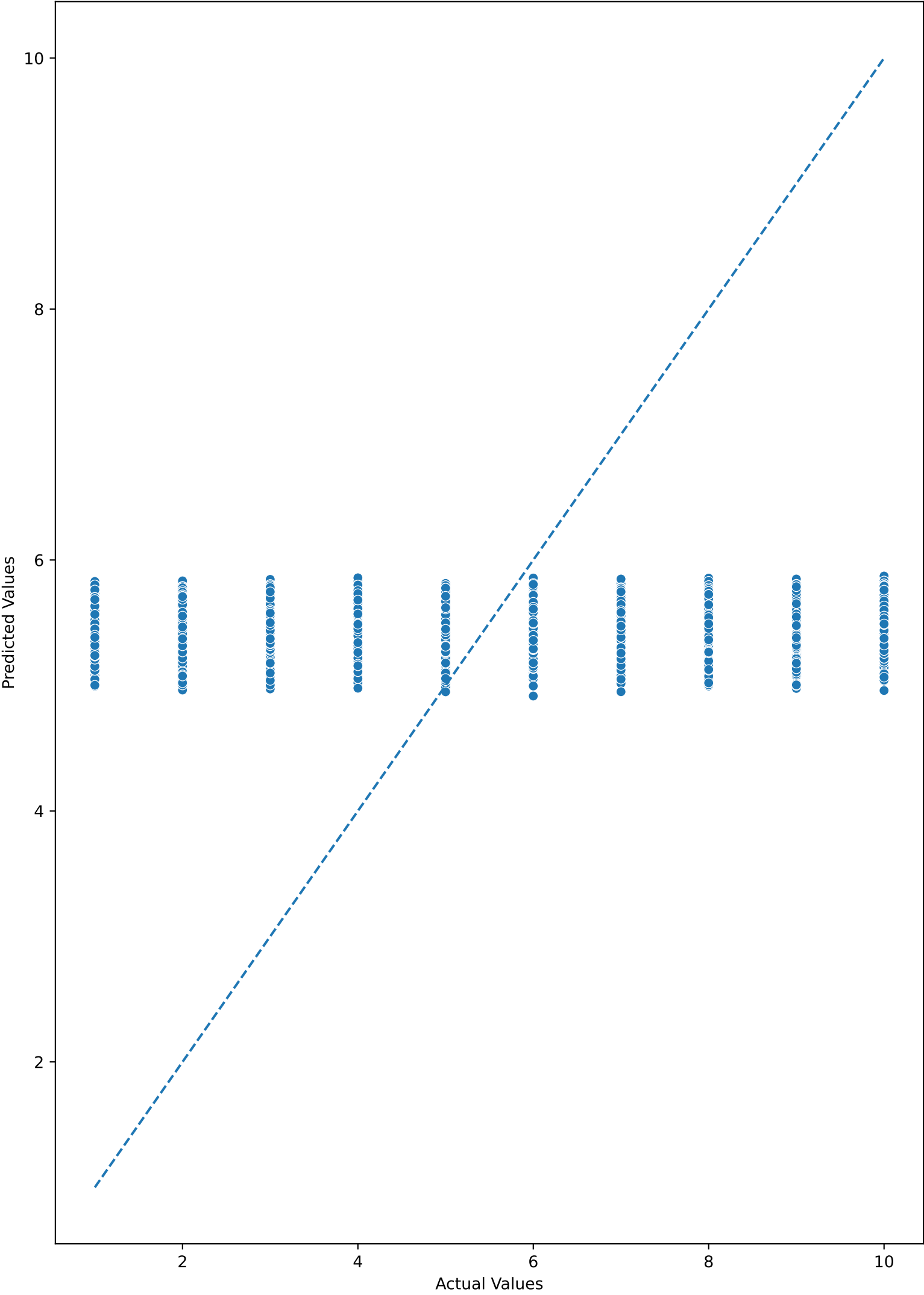
RMSE Comparison



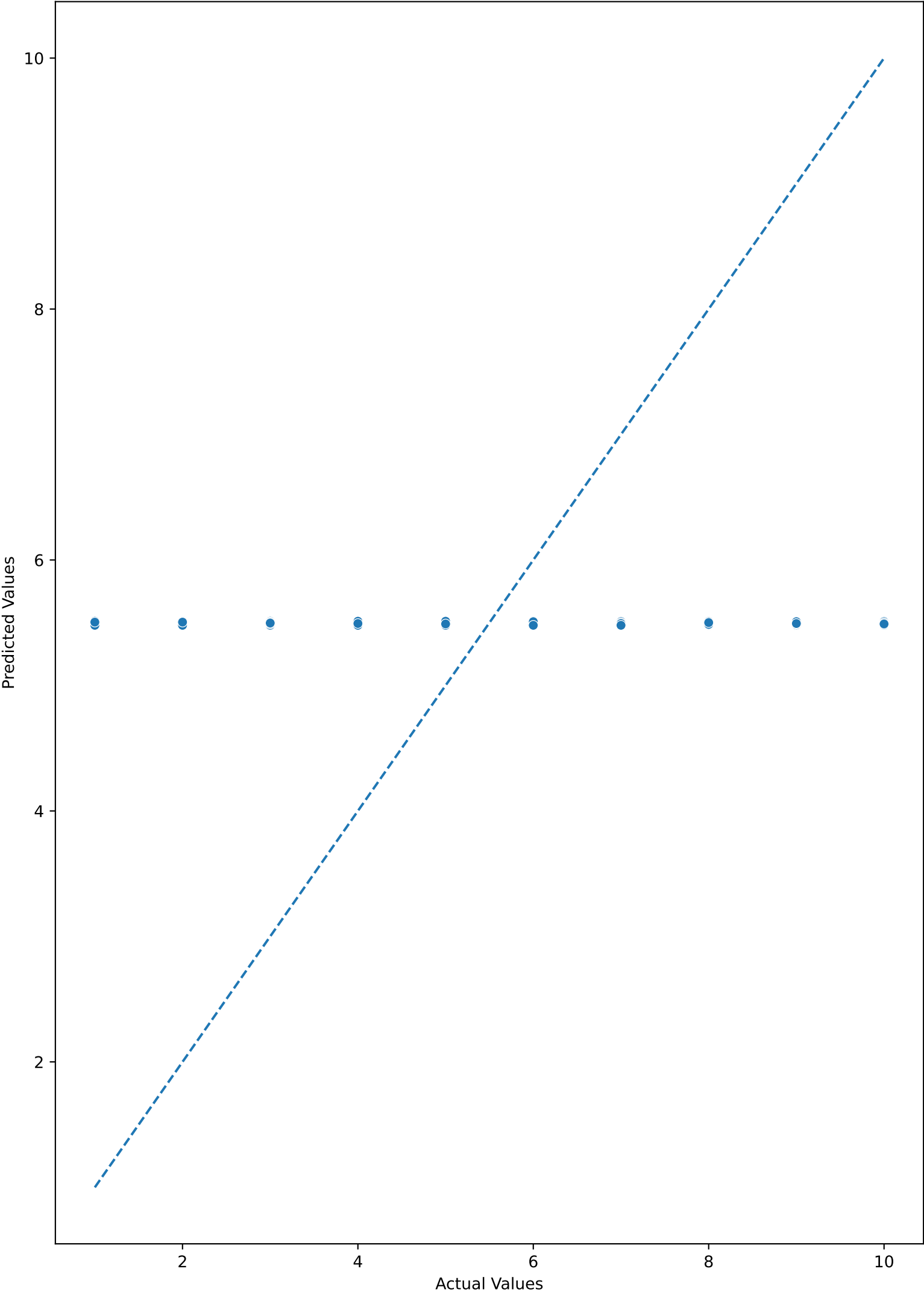
Actual vs Predicted - Linear Regression



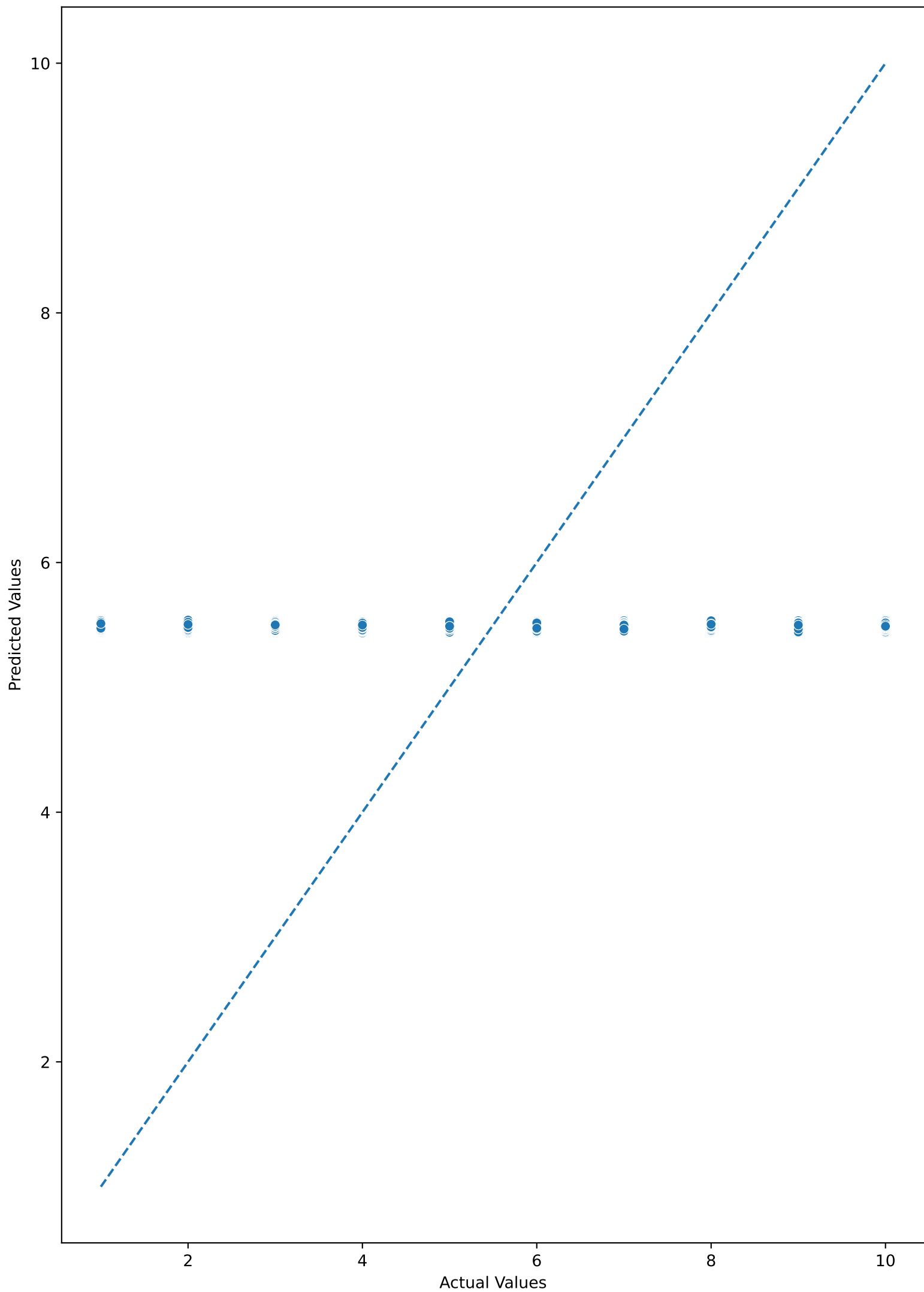
Actual vs Predicted - Ridge Regression



Actual vs Predicted - Lasso Regression



Actual vs Predicted - Elastic Net



FINAL OBSERVATIONS

Best Model: Lasso Regression

Best R2 Score: -0.0000

Best RMSE: 2.8707

Lasso Zero Coefficients: 158

Elastic Net Zero Coefficients: 152

Ridge Regression applies L2 regularization.

Lasso Regression applies L1 regularization.

Elastic Net combines L1 and L2 regularization.